

If the entire world was to switch over to LEDs, there would be tremendous savings in energy costs and an increase in energy security. In addition to their high efficiency and long lifespan, LEDs do not require ballast and do not contain toxic mercury, which means disposing them of poses fewer health and environmental hazards.

Growing applications

In contrast to conventional light sources, a number of new dimensions have opened due to the unique controllability of LEDs. These include control over the emission spectrum, colour temperature, polarisation, temporal modulation, hue and spatial emission pattern. These controllable next-generation LED sources, called smart lightings, will result in tremendous benefits to the society and humankind. In general, applications of LEDs in different walks of life include:

Biology and imaging. Leapfrog advances in quantitative biology, particularly the rapid identification and counting of biological cells through adaptive and fully-tunable reflectance and fluorescence imaging.

Display systems. Liquid-crystal-displays and projectors with unprecedented efficiency and brilliancy (huge colour gamut) through polarisation-controlled lighting sources.

Transportation. Enhanced visibility (less glare) and safety through polarisation-controlled headlights, temporal-controlled communicating headlights/brake lights/traffic lights, and interactive roadways.

Communications. Fundamentally new modes of broadcasting, communications and sensing through temporal control of solid-state light sources.

Human factors. Reduced dependency on sleep-inducing pharmaceuticals, prevention of certain cancers and higher quality of life.

Agriculture. Efficient plant growth in non-native regions (including space), non-native seasons and revolutionised indoor agriculture and urban indoor vertical farming or sky farming.

Limitations

So far, despite the positive forecast, there remain short-term financial and technological obstacles to the market success of LEDs, with the biggest problem being price. LED tubes cost anywhere between \$50 and \$100, compared to \$2-\$10 for fluorescent.

The other roadblock frequently cited by electrical engineers and facilities managers is brightness; they are concerned that occupants will complain that the lights are not bright enough. Both problems are being addressed by LED manufacturers, who are researching ways to increase the luminance of LEDs and cut costs.

Recent developments

Researchers have developed and demonstrated a new type of LED with significantly improved lighting performance and energy efficiency. The new polarisation-matched LED exhibits an 18 per cent increase in light output and a 22 per cent increase in wall-plug efficiency, which essentially measures the amount of electricity that the LED converts into light. The new device achieves a notable reduction in 'efficiency-droop'—a phenomenon that provokes LEDs to be most efficient when receiving low-density currents of electricity, but then lose efficiency as higher density currents of electricity are fed into the device. The cause of this droop is not yet fully understood, but studies have shown that electron-leakage is likely a large factor for the problem.

This droop is under the spotlight since today's high-brightness LEDs are operated at current densities far beyond efficiency-peaks. This challenge has been a stumbling block, because reducing the current densities to values where LEDs are more efficient is unacceptable. However, new LEDs, which have a radically re-designed polarisation-matched active region, tackles this issue and brings LEDs closer to being able to operate efficiently at high current densities.

Focusing on the active region

of LEDs, where the light is generated, researchers discovered that the region contained materials with mismatched polarisation. The polarisation-mismatch possibly causes electron leakage, and therefore a loss in efficiency. The researchers discovered that the polarisation-mismatch can be reduced by introducing a new quantum-barrier design. They replaced the conventional gallium indium nitride/gallium nitride (GaInN/GaN) layer of the LED active region with GaInN/GaN. This substitution allows the layers of the active region to have a better-matched polarisation, resulting in reduced electron-leakage and efficiency-droop.

Although reports suggest that paying more for LEDs would be worth it, as long-term use of LEDs will compensate their operational cost as compared to fluorescent bulbs, still, as with most green products, the long-term cost savings of LEDs won't necessarily convince users right away. It's likely to take a while for them to warm up to the idea that they should increase their lighting budget by a few thousand dollars, even if it means reducing their energy bills. Researchers at Purdue University found a way to replace the expensive standard substrate used in LED production—sapphire—with low-cost, metal-coated silicon wafers. The switch would greatly reduce the cost of LED manufacturing, assuming another silicon shortage does not come along.

The way forward

Along with much of the attention being focused on the next-generation energy-efficient lighting—solid-state technologies like LEDs with their vast household as well as industrial applications—companies have also been working on new LED fixtures and bulbs that promise to cut overall operational lighting costs by up to 70 per cent. A study predicted that it will take ten years for LEDs to overcome all types of obstacles to their mass adoption. ●

The author is an associate professor in Department of Physics, S.L.I.E.T., Longowal, District Sangrur, Punjab